

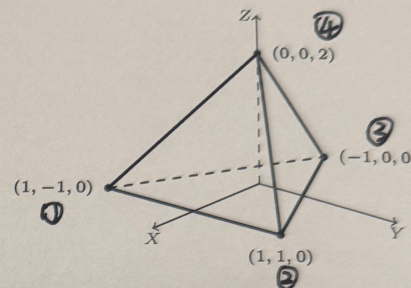
Quiz 1

Thursday, September 3, 2026

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Closed book and laptop. Keep each answer to a line or two. (10 points total)

1. (2 pts) The triangular pyramid on the right is a triangle mesh. Write down a vertex matrix V and a face matrix F for it — give the size of each, and use any vertex numbering you like as long as F is consistent with it.



$$V = \begin{bmatrix} 1, -1, 0 \\ 1, 1, 0 \\ -1, 0, 0 \\ 0, 0, 2 \end{bmatrix}$$

$$F = \begin{bmatrix} 1, 1, 0, 1 \\ 0, 1, 1, 1 \\ 1, 1, 1, 0 \\ 1, 0, 1, 1 \end{bmatrix}$$

2. (4 pts) $f_1(x)$ and $f_2(x)$ are implicit functions for two shapes (negative inside, positive outside). Write an implicit function for the set of points that lie (i) inside either shape, and (ii) inside the second shape but not inside the first.

(i) inside either shape

$$\{p \mid f_1(p) < 0 \vee f_2(p) < 0\}$$

(ii) inside the second shape not inside the first:

$$\{p \mid f_2(p) < 0 \wedge f_1(p) \geq 0\}$$

3. (4 pts) This pseudocode is supposed to sample N points uniformly over the surface of a mesh, but it does not. Circle the line(s) that are wrong and write the fix.

samplePoints(V, F, N):for $i = 1 \dots N$: $f = \text{uniformRandomChoice}(F)$ $(a, b, c) = V[f]$ $u = \text{rand}(0, 1); v = \text{rand}(0, 1)$ $P[i] = u*a + (1-u)*v*b + (1-u)*(1-v)*c$ return P

$f = \text{SampleSurfaceBasedOnSurfaceSize}(F)$
(Proportional to)